

Instructions

- Answer all eight questions. Show sufficient working for method marks.
- A calculator may be used unless a question states otherwise.
- Write clearly in the spaces provided. Additional pages may be attached.

SCORE SAVED TO COURSE PROGRESS

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1

Evaluate the limit and justify any algebraic simplification.

[4 marks]

$$\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3}$$

Method sound; final simplification omitted.

4/4

2

Differentiate the function and simplify your answer.

[5 marks]

$$y = (3x^2 + 1)^5$$

Check the condition before applying the rule.

5/5

AGENT ANNOTATED

3

A rectangle has perimeter 40 cm. Find the dimensions that maximise its area, and show your result is a maximum.

[6 marks]

Correct and clearly justified.

5/6

4

Evaluate the definite integral, showing the antiderivative you use.

[5 marks]

$$\int_0^2 (3x^2 - 4x + 1) dx$$

Method sound; final simplification omitted.

4/5

5

Find the equation of the tangent to the curve below at the point where $x = 2$.

[5 marks]

$$y = x^3 - 2x$$

Check the condition before applying the rule.

5/5

AGENT ANNOTATED

6

Solve the differential equation given that $y(0)=2$.

[6 marks]

$$\frac{dy}{dx} = 3y$$

Correct and clearly justified.

5/6

7

Decompose into partial fractions, stating the form you assume before finding the coefficients.

[6 marks]

$$\frac{5x + 1}{(x - 1)(x + 2)}$$

Method sound; final simplification omitted.

6/6

8

Integrate by parts, stating your choice of u and dv .

[5 marks]

$$\int x e^x dx$$

Check the condition before applying the rule.

4/5